

Jude Lwin

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Education

University of Maryland, College Park

Bachelor of Science, Computer Science

GPA: 3.9/4.0

May 2027

Technical Skills

Languages: Python, C++, Java, TypeScript, JavaScript, C, Rust, OCaml, Go, SQL, R

Frameworks/Libraries: Scikit-Learn, PyTorch, Pandas, FastAPI, Flask, React, Node.js, Express

Technologies/Tools: Git, Firebase, AWS, Docker, PostgreSQL, Snowflake, MongoDB, Unix, CI/CD, REST APIs

Experience

Amazon

New York, NY

Software Engineer Intern

Feb. 2026 – Present

- Extended authorization infrastructure (41K+ applications, 198K+ service relationships) to secure **agentic AI** by building a **hook**-based runtime enforcement layer to prevent unsafe behavior caused by prompt injection
- Designed a deterministic, session-level **policy engine** that tracks cumulative risk and blocks unsafe multi-step tool combinations across MCP-based workflows
- Built **Java** backend on **AWS Lambda** exposing 12 tools via **MCP**, integrating with **OAuth**-based authorization
- Executed end-to-end attack simulations across 4 scenarios on an onboarded agent, demonstrating human-in-the-loop approval for unsafe actions and full **MCP**-to-auth execution flow

Capital One

McLean, VA

Software Engineer Intern

Jun. 2025 – Aug. 2025

- Built LLM evaluation pipeline for **RAG** chatbot in **Python/Snowflake**, scaling model validation from dozens to 10K+ queries per deployment, reducing engineering time from days of manual review to overnight batch processing
- Trained ensemble classifiers with **scikit-learn** (Random Forest, XGBoost) and developed **NLP** scoring system (ROUGE, Levenshtein, fuzzy encoding) achieving 91.2% accuracy in deployment verification
- Engineered backtesting framework to validate metric/classifier updates against all previous evaluation runs

University of Maryland

College Park, MD

Machine Learning Researcher | Advisor: Dr. Furong Huang

Sep. 2025 – Present

- Built **Python** evaluation pipeline for reasoning self-improvement, ACL first author submission under review
- Implemented similarity search tooling over DeepMath-103K to inject question/answer pairs into model context

WISE Cities

College Park, MD

Software Engineer Intern | AI/ML Club

Feb. 2025 – May 2025

- Led backend development of a recommendation system for older adults to find local organizations and activities
- Engineered scalable **gRPC** microservice with **MongoDB** querying and **Pydantic** data validation
- Integrated all-MiniLM-L6-v2 and **Pinecone** to enable sub-100ms k-NN recommendations

University of Maryland

College Park, MD

Teaching Assistant, CMSC330 (Programming Languages)

Jan. 2025 – Jan. 2026

- Led weekly discussion for 30+ students, hosted 5+ weekly office hours, and coordinated grading with 39 TAs
- Developed and tested 7 **OCaml/Rust** projects, wrote 50+ unit tests and documentation for 900+ students
- Streamlined releases using GitHub, **Docker**, and GitHub Actions, improving deployment efficiency

Projects

Personalized Restaurant Recommender | *Python, PyTorch, FastAPI, React, TypeScript, PostgreSQL*

- Built a two-stage restaurant recommendation system with a **PyTorch** neural ranker to surface quick, affordable nearby meals using user, restaurant, and contextual features
- Engineered an ML training pipeline with simulated interaction logs and offline evaluation, achieving **90%+** classification accuracy across **100+** real Manhattan restaurants

Distributed Build System 🦄 | *C++, gRPC, Redis, Docker*

- Built a distributed build system that parallelizes **C++** compilation tasks across networked nodes using **gRPC** coordination and dependency-graph scheduling
- Added artifact caching with **Redis** and containerized benchmarks to evaluate scalability and fault recovery

Smart Study Assistant 🦄 | *Python, FastAPI, Celery, Docker, Supabase, Pinecone, AWS S3, React*

- Developed a distributed study assistant using **FastAPI**, **React**, and **Docker** microservices, allowing students to organize notes by course and query them in natural language to retrieve cited excerpts for efficient review
- Implemented RAG pipeline (**Celery**, **Pinecone**) for PDF parsing, chunking, and sub-200ms semantic search